Status	Finished
Started	Friday, 16 October 2026, 4:05 PM
Completed	Friday, 16 October 2026, 4:15 PM
Duration	10 mins 28 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The first element is returned
b. The last element is returned
c. An error is raised
d. The search is considered unsuccessful ✓

Question **2**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the middle of the list
b. In the right sublist ✓
c. In the left sublist
d. At the beginning of the list

Question **3**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(n log n)
b. O👎 ✓
c. O(log n)
d. O(1)

Question **4**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It works on unsorted lists
b. It always starts from the beginning of the list
c. It compares elements sequentially
d. It can be applied only if the list is sorted ✓

Question **5**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(1) ✓
b. O(log n)
c. O👎
d. O(n log n)

Question **6**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops
b. The list is sorted
c. The search continues from the beginning
d. The search continues in the left or right sublist ✓

Question **7**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search has a lower time complexity on large, sorted lists ✓
b. Binary search does not require dividing the list
c. Binary search can find multiple instances of the target element
d. Binary search works on unsorted lists

Question **8**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 90 and 94

b. 90 and 99 ✓

c. 89 and 94

d. 89 and 99

Question **9**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is small and unsorted ✓

b. When the list is large and unsorted

c. When the list is large and sorted

d. When the list is small and sorted

Question **10**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Divide the list into two equal parts

b. Compare the target element with the first element in the list

c. Compare the target element with the middle element in the list ✓

d. Sort the list

Question **11**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. log n
b. n/2
c. n ✓
d. n-1

Question **12**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search stops immediately ✓
b. The search backtracks to find duplicate elements
c. The search continues until the end of the list
d. The search starts over from the beginning

Question **13**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By starting from the first element
b. By comparing each element sequentially
c. By using a hash function
d. By dividing the list length by two ✓

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Dividing the list in half repeatedly
b. Skipping every second element
c. Checking each element sequentially ✓
d. Sorting the list before searching

Question **15**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Finding

b. Discovering

c. Searching ✓

d. Mining